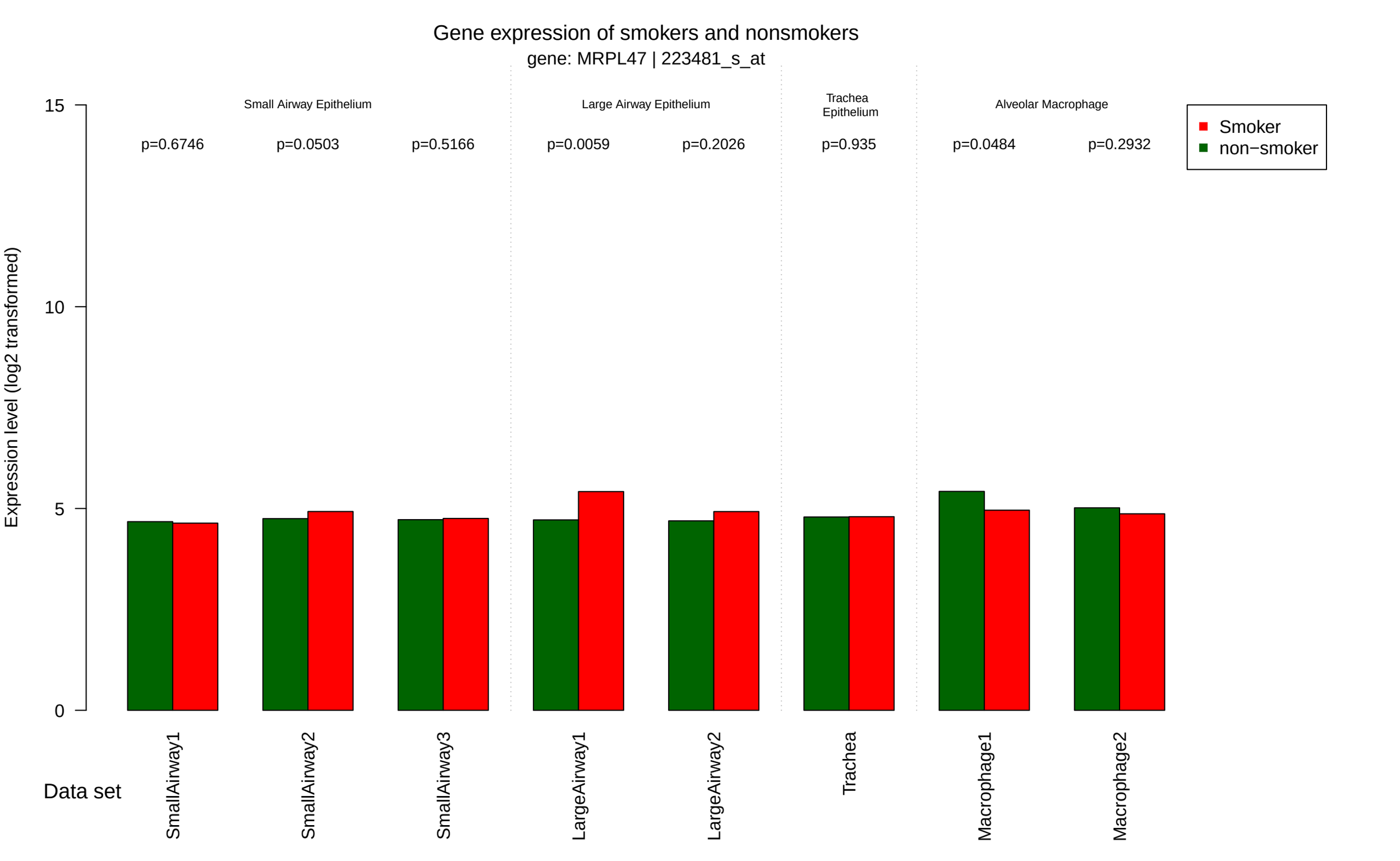
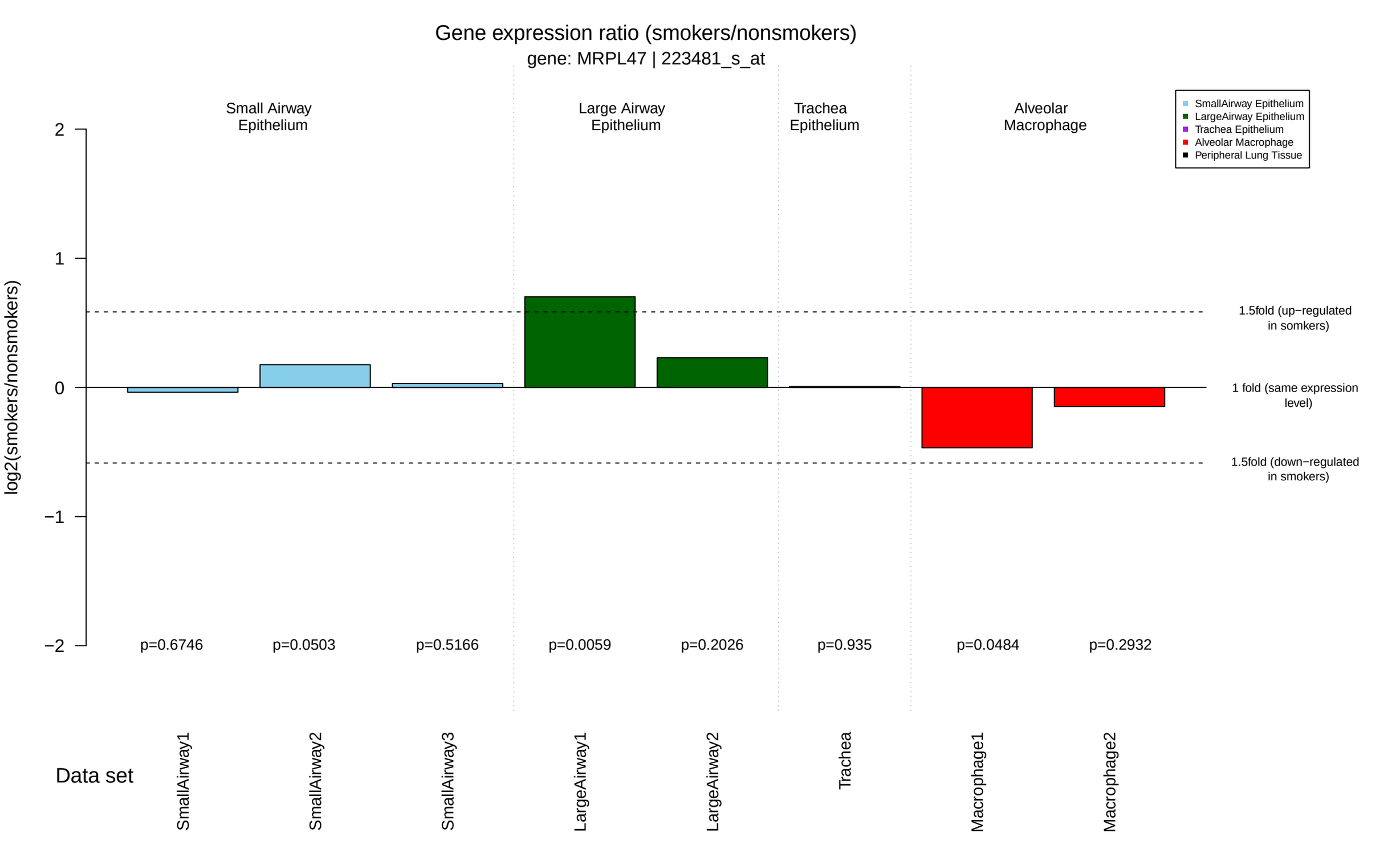
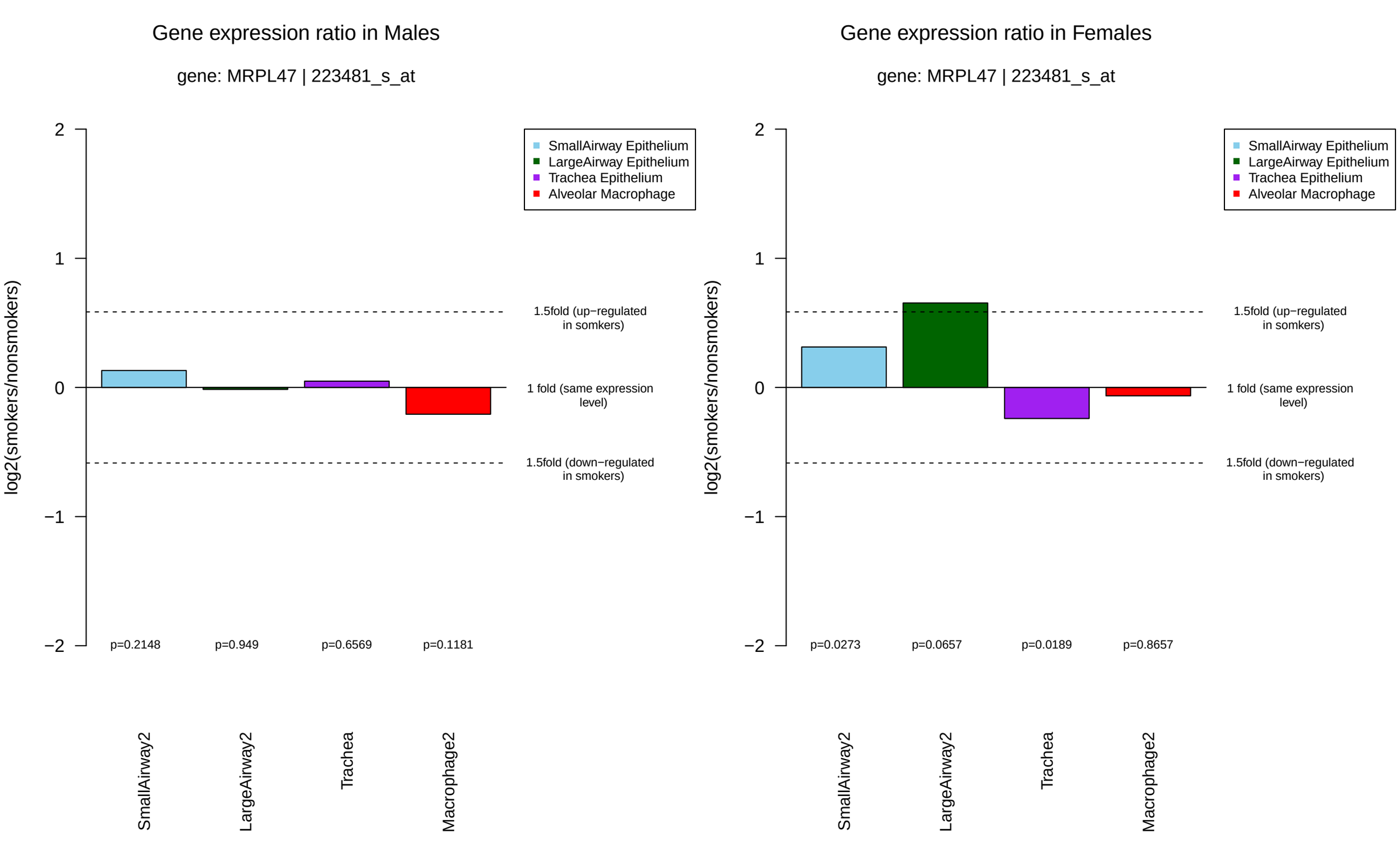
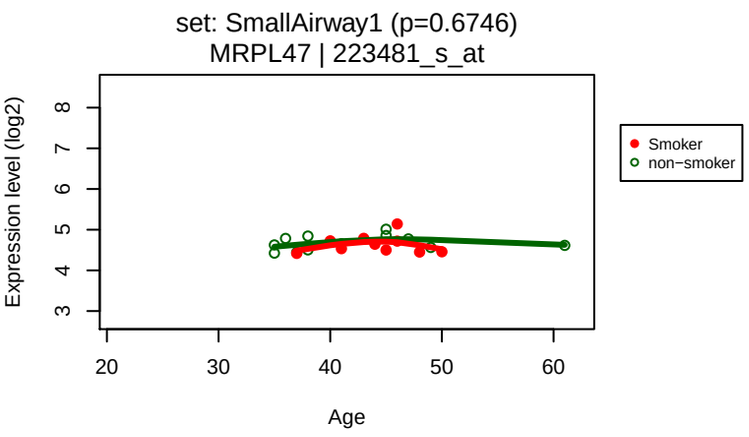
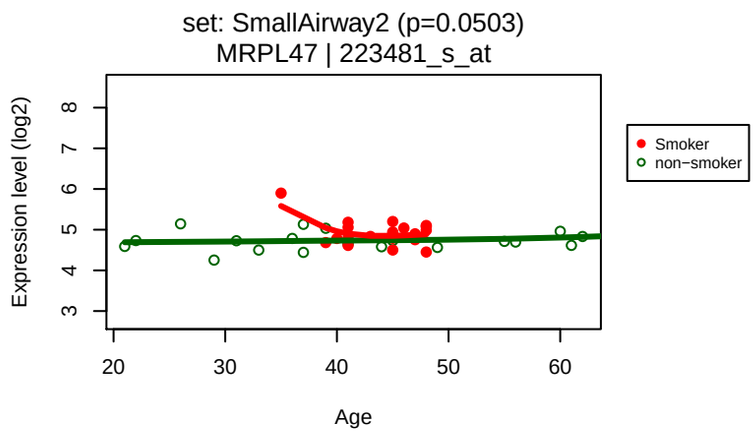
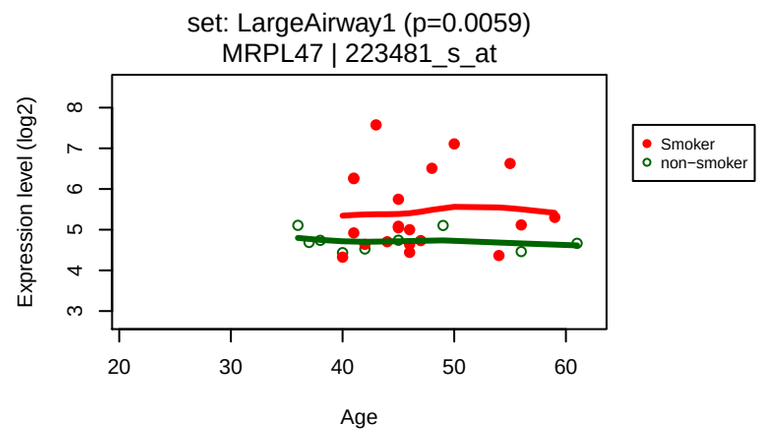

Gene expression of smokers and nonsmokers
gene: MRPL47 | 223481_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6746
p=0.0503
p=0.5166
p=0.0059
p=0.2026
p=0.935
p=0.0484
p=0.2932
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

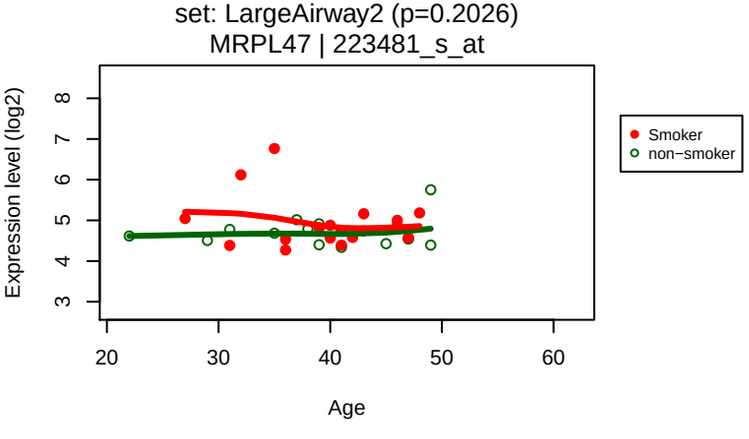

Gene expression ratio (smokers/nonsmokers)
gene: MRPL47 | 223481_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6746
p=0.0503
p=0.5166
p=0.0059
p=0.2026
p=0.935
p=0.0484
p=0.2932
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

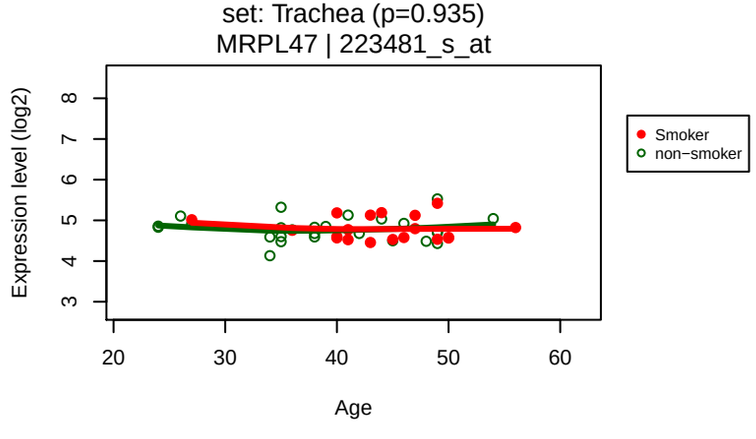

Gene expression ratio in Males
gene: MRPL47 | 223481_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2148
p=0.949
p=0.6569
p=0.1181
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL47 | 223481_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0273
p=0.0657
p=0.0189
p=0.8657
SmallAirway2
LargeAirway2
Trachea
Macrophage2

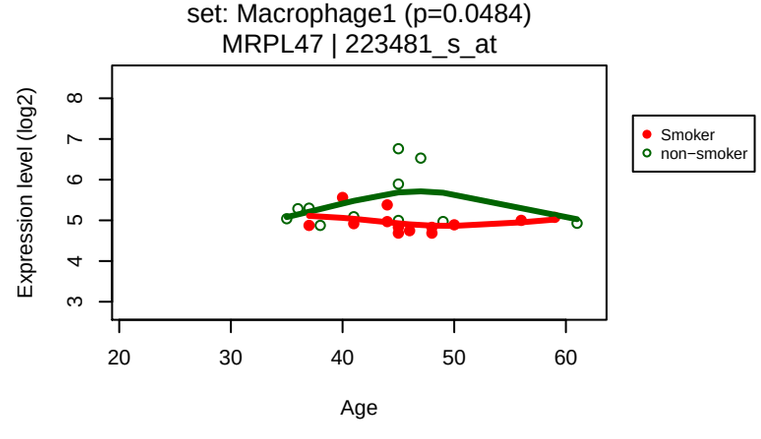

set: SmallAirway1 (p=0.6746)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker

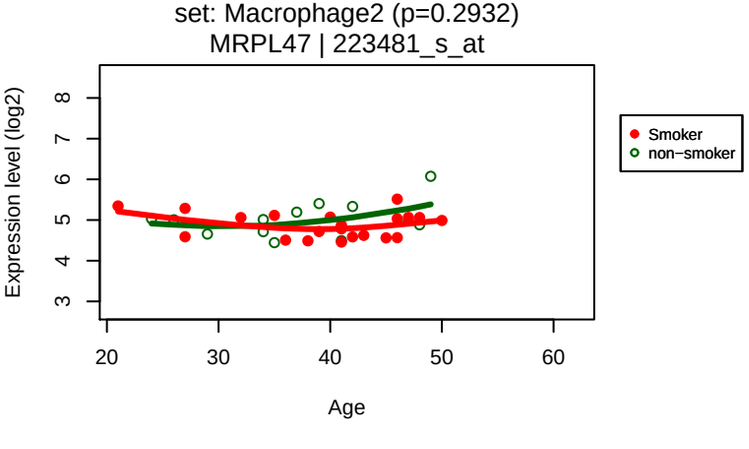

set: SmallAirway2 (p=0.0503)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker

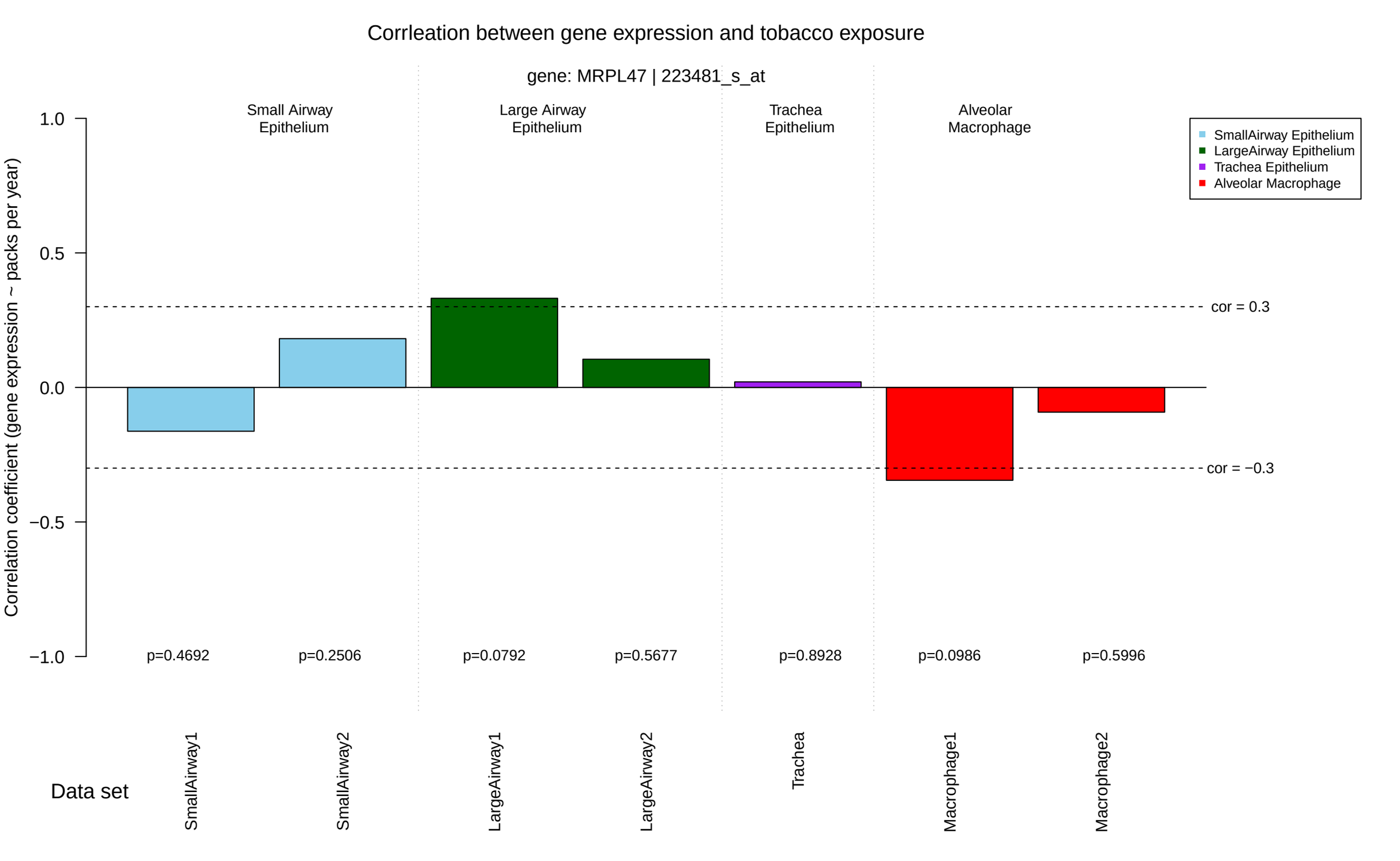

set: LargeAirway1 (p=0.0059)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2026)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.935)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0484)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2932)
MRPL47 | 223481_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL47 | 223481_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4692
p=0.2506
p=0.0792
p=0.5677
p=0.8928
p=0.0986
p=0.5996
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2